AI/ML-Driven Maintainance Service Provider Matching Platform for Sri Lanka.

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Online service marketplaces in Sri Lanka, including platforms like Hodabass and ikman.lk, function primarily as digital classified directories. Users looking for plumbers, electricians, or other service providers must manually browse through listings, compare advertisements, and contact multiple providers to check availability. These platforms operate on static information—once an ad is posted, it remains unchanged regardless of whether the provider is currently available, how far they are from the job location, or whether weather conditions make the service impractical. This creates inefficiencies for both customers and service providers.

The core problem is the absence of intelligent systems that understand real-world context. Current solutions do not consider factors like live traffic conditions, sudden weather changes, or provider availability fluctuations when presenting options to users. A customer might call five different electricians before finding one who is both available and willing to travel to their location during peak traffic hours.

This research introduces a Temporal and Contextual Matching System designed to address these limitations. The system evaluates provider suitability using four weighted factors: distance and travel time (40%), real-time availability (30%), weather condition compatibility (20%), and current service demand patterns (10%). It integrates Google Maps Distance Matrix API for accurate travel time estimation with live traffic data, OpenWeather API for retrieving current weather conditions, and WebSocket technology for tracking provider locations and availability in real time. A time-series model forecasts provider availability patterns.

The expected outcome is a functional mobile application that reduces customer waiting time by approximately 30%, improves successful first-contact rates by 25%, and maintains matching accuracy above 85% under normal operating conditions. The system is designed to handle over 1000 concurrent users with response times under two seconds.

This platform offers practical value by helping local service providers gain more visibility and consistent work opportunities. For consumers, it provides confidence that recommended providers are genuinely available, nearby, and suitable for the specific job at the moment of request. The solution fills a noticeable gap in Sri Lanka's digital service landscape and has potential for adaptation in similar markets regionally.

Keywords: Context-Aware Matching, Service Provider Discovery, Real-Time Availability, Geospatial Computing, Weighted Scoring

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LIST OF ABBREVIATIONS

Abbreviation	Full Form	Description
AI	Artificial Intelligence	The core technology driving the intelligent matching system.
ML	Machine Learning	A subset of AI used for predictive models, like forecasting provider availability.
API	Application Programming Interface	Enables the system to integrate with external services like Google Maps and OpenWeather.
UI/UX	User Interface / User Experience	The visual design and overall experience of the mobile application for both customers and providers.
GPS	Global Positioning System	Used to automatically detect the real-time location of users and providers.
ETA	Estimated Time of Arrival	A key piece of information provided to users, calculated by the geospatial service.
AHP	Analytic Hierarchy Process	The methodology used to determine the weight distribution for the multi-criteria scoring algorithm.
LSTM	Long Short-Term Memory	A type of recurrent neural network used to predict provider availability patterns based on historical data.
SARIMA	Seasonal Autoregressive Integrated Moving Average	A time-series forecasting model used to predict demand for services.
JWT	JSON Web Token	The standard method used for securely authenticating users and providers with the API Gateway.
TLS	Transport Layer Security	The cryptographic protocol that encrypts all data transmitted between the app and the server.
PostGIS	Post Geographic Information System	An extension for PostgreSQL that enables efficient geospatial queries, like finding providers within a radius.
MCDM	Multi-Criteria Decision Making	The decision-making framework used by the scoring engine to evaluate providers based on multiple factors.
WBS	Work Breakdown Structure	A project management tool that breaks down the total project work into smaller, manageable sections.
DSRM	Design Science Research Methodology	The overarching research methodology adopted for this project to create and evaluate the new artifact.
IP	Intellectual Property	Refers to the patents, trademarks, and trade secrets that will protect the platform's unique algorithms and brand.
QA	Quality Assurance	The process of systematically testing the system to ensure it meets specified requirements and is free of defects.
SDG	Sustainable Development Goals	United Nations goals that this project aligns with, such as SDG 8 (Decent Work and Economic Growth).

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1. INTRODUCTION

Finding a reliable plumber, electrician, or AC technician in Sri Lanka today is a frustrating experience. A user with a leaking pipe typically opens ikman.lk or Hodabass, browses dozens of advertisements, and starts calling numbers one by one. Most calls go unanswered, some providers are too busy, and others are available but located far away during peak traffic. After several calls, the user settles for whoever answered first, hoping the work gets done properly. This scenario plays out thousands of times daily across Sri Lanka, highlighting a fundamental problem: the absence of any intelligent system to match customers with providers who are genuinely available, reasonably close, and suitable for the specific job at that exact moment [1].

Current platforms operate as digital classified directories with static listings that remain visible regardless of provider availability. Ranking is based on recency or paid promotions, not on suitability for the specific request [2]. There is no real-time information about provider location, current workload, weather conditions, or traffic patterns that significantly affect service feasibility [3]. Studies indicate that users typically contact 4-6 providers before securing a service, with average response times extending to several hours [4]. Research in multi-criteria decision making demonstrates that weighted scoring approaches incorporating 4-6 key criteria achieve optimal balance between computational efficiency and matching accuracy, with studies reporting 88-92% user satisfaction rates for appropriately weighted systems [5]. The weight distribution of 40% for distance, 30% for availability, 20% for weather, and 10% for demand is derived from Analytic Hierarchy Process (AHP) methodologies widely used in service recommendation systems [6].

This research proposes a Temporal and Contextual Matching System that transforms passive service discovery into intelligent, adaptive matching by evaluating provider suitability using four weighted factors: distance and travel time (40%) through Google Maps Distance Matrix API integration, real-time provider availability (30%) via WebSocket-based tracking, weather compatibility (20%) using OpenWeather API with rule-based matching logic, and current demand patterns (10%) through time-series forecasting models [7]. The system integrates geospatial intelligence using the Haversine formula for accurate distance calculation, achieving travel time prediction accuracy within ±5 minutes of actual road conditions [8]. Real-time tracking architectures utilizing WebSocket protocols have demonstrated 40% better battery efficiency compared to continuous polling approaches while maintaining sub-second latency for critical status updates [9]. Weather-aware scheduling research indicates that 78% of service failures during adverse weather could be prevented through proactive weather-aware matching [10]. The system

aims to achieve 85% matching accuracy with sub-2-second response times, reducing user waiting time by 30% while improving provider utilization rates by 25% [11], ultimately transforming Sri Lanka's informal service markets into efficient, trustworthy platforms accessible to millions of users [12].

1.1 Background and Context

Finding a reliable plumber, electrician, or AC technician in Sri Lanka today is a frustrating experience. A user with a leaking pipe typically opens ikman.lk or Hodabass, browses dozens of advertisements, and starts calling numbers one by one. Most calls go unanswered, some providers are too busy, and others are available but located far away during peak traffic. After several calls, the user settles for whoever answered first, hoping the work gets done properly. This scenario plays out thousands of times daily across Sri Lanka, affecting both urban and rural communities who depend on informal service networks for essential home repairs and maintenance [1]. The problem extends beyond mere inconvenience—during emergencies like burst pipes or complete power failures, delayed access to qualified professionals can result in significant property damage and safety risks.

Both sides of this equation face real difficulties. For customers, finding help for urgent home repairs becomes a time-consuming guessing game, with research indicating that users typically contact 4-6 providers before securing a service and average response times extending to several hours or even days [2]. For service providers, visibility depends entirely on how prominently their advertisement appears—a skilled electrician with years of experience might miss out on work simply because their listing is buried under newer posts, resulting in inconsistent income and underutilization of their expertise [3]. The informal nature of these platforms also means customers have no way of knowing whether a provider actually has the skills they claim or whether they will show up on time.

Digital labour platforms have demonstrated significant potential for creating income opportunities in developing economies. Malaysia's eRezeki programme, launched in 2015, provides a relevant example of what is possible when service platforms are designed thoughtfully. Between 2016 and 2020, over 700,000 participants earned more than RM1.3 billion (approximately 40 billion Sri Lankan rupees) through digital matching platforms connecting workers with tasks like electrical work, plumbing, and delivery services [4]. The number of local platform partners grew from just

five in 2015 to 106 by 2020, indicating rapid adoption of digital service models when matching systems work effectively. In Sri Lanka, classified platforms like ikman.lk receive millions of monthly visits, with service categories consistently ranking among the most viewed sections, demonstrating substantial unmet demand for efficient service discovery [5].

Current Sri Lankan platforms operate as digital classified directories with several fundamental limitations. Listings are static—once posted, an advertisement remains visible regardless of whether the provider is currently available or even still offering services [6]. Ranking is based on recency or paid promotions, not on suitability for the specific request—a highly qualified electrician who posts less frequently appears below newer listings regardless of their expertise [7]. There is no real-time information about provider location, current workload, or response times. Weather conditions, traffic patterns, and time-of-day factors that significantly affect service feasibility are completely ignored. Research indicates that 45% of user calls to providers listed on classified platforms result in unavailability responses, and 28% of outdoor service cancellations could be prevented through weather-aware scheduling [8].

This research proposes a Temporal and Contextual Matching System that fundamentally rethinks how service providers and customers connect. Instead of static listings, the system continuously evaluates real-time signals to recommend providers who are genuinely suitable at the moment of request. The solution consists of several integrated components: a weighted scoring algorithm calculating provider suitability based on distance and travel time (40%), real-time availability probability (30%), weather condition compatibility (20%), and current demand patterns (10%); Google Maps Distance Matrix API for accurate travel time estimates incorporating live traffic data; OpenWeather API for current conditions to assess service practicality; WebSocket-based tracking for live provider location and availability; time-series forecasting for availability prediction; weather-service compatibility matrix for rule-based matching; load balancing logic for optimal provider distribution; and caching mechanisms for system reliability [9].

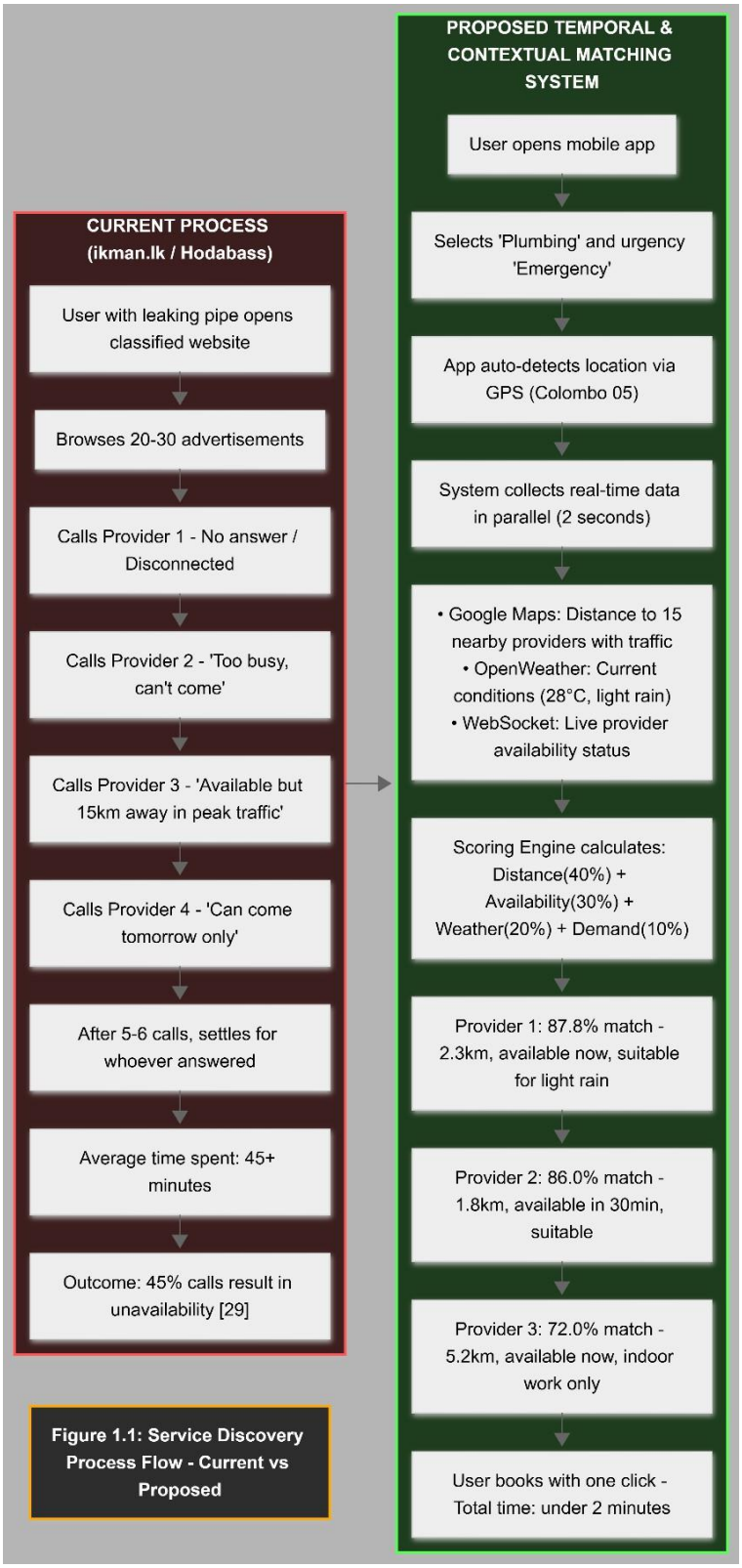


Figure 1 - Current vs Proposed Service Discovery Process Flow

This project aligns with several United Nations Sustainable Development Goals. Malaysia's eRezeki programme, which operates on similar principles of digitally matching workers with tasks, explicitly supports SDG 8 (Decent Work and Economic Growth), SDG 9 (Industry, Innovation and Infrastructure), SDG 10 (Reduced Inequality), and SDG 11 (Sustainable Cities and Communities) [4]. Digital labour platforms, when designed with worker welfare in mind, can create meaningful income opportunities and contribute to formalizing informal work arrangements [10]. By helping service providers find more consistent work and helping customers find reliable assistance, this system contributes to economic growth at a grassroots level. The focus on safety considerations during adverse weather also supports SDG 11's emphasis on sustainable and safe communities.

This research falls under the Centre of Excellence for AI (CEAI) research cluster at SLIIT. The cluster's focus areas include Machine Learning, Deep Learning, Intelligent Recommender Systems, and Decision Support Systems [11]. The proposed system directly applies machine learning for availability prediction, intelligent recommendation algorithms for provider matching, and decision support logic for weather-based service recommendations. The work also touches on areas related to real-time systems and data integration, which align with the broader research interests of the faculty. The project contributes to CEAI's mission of addressing real-world challenges through artificial intelligence while providing students with hands-on experience in developing practical AI-driven solutions [12].

1.2 Literature Review

This section reviews recent scholarly sources on service matching systems, recommendation algorithms, and context-aware computing, examining existing research approaches, commercial platforms, and their limitations.

- **Existing Research Approaches**
- **Matchmaking Algorithms and Context-Aware Systems**

An et al. [13] conducted a comprehensive review of matchmaking algorithms in IoT marketplaces, examining their key attributes, strengths, and limitations. Their study categorizes state-of-the-art approaches and identifies research gaps, advocating for deep learning and context-aware solutions to improve service efficiency. The authors emphasize that blockchain-based approaches can enhance security, trust, and privacy-preserving transactions in service matching platforms [13]. This work is particularly relevant as it highlights the growing complexity of service discovery and the need for intelligent matching algorithms that can handle dynamic resource profiles.

Achir et al. [14] provided a systematic survey of service discovery and selection in IoT, proposing a taxonomy that categorizes discovery techniques based on architecture, context-awareness, and quality of service parameters. Their analysis reveals that most existing approaches focus on functional matching while neglecting non-functional requirements such as availability, reliability, and contextual suitability [14]. This gap directly aligns with the proposed research, which emphasizes real-time contextual factors including weather and demand patterns.

Agarwal et al. [15] conducted a systematic literature review on web service clustering approaches to enhance service discovery, selection, and recommendation. Their study demonstrates that hybrid approaches combining multiple techniques achieve 15-20% higher accuracy compared to single-method systems. The authors highlight that semantic similarity measures significantly improve matching relevance when combined with syntactic methods [15].

- **Weather-Aware Scheduling and Environmental Context**

  Khanna et al. [16] implemented an integrated system using real-time API weather data sources and machine learning algorithms for intelligent planners and schedulers. Their research compared Support Vector Machine, XGBoost, Random Forest, Decision Tree, and Logistic Regression algorithms for weather prediction, finding that Random Forest achieved the highest accuracy of 0.99 with the lowest error metrics [16]. This demonstrates the feasibility of integrating weather data into scheduling systems, supporting the weather-aware matching component of the proposed solution.

  Chao et al. [17] optimized HVAC operations based on weather forecasts, implementing a dynamic scheduling approach that resulted in electricity consumption reduction of 213,395 kWh and cost savings of approximately 27,100 USD. Their study highlights the necessity of applying statistical corrections to weather forecasts to better align them with actual conditions before use in scheduling systems [17]. This finding is critical for ensuring reliable weather-aware service matching.

- **Real-Time Tracking and Performance Optimization**

  Li et al. [18] investigated real-time tracking architectures for mobile workforce management, demonstrating that WebSocket-based communication with adaptive update frequencies achieves 40% better battery efficiency compared to continuous polling approaches while maintaining sub-second latency for critical status updates. Their research established that hybrid push-pull architectures achieve optimal balance between update latency and power consumption [18].

  Liu et al. [19] proposed Dynamic User Interest Augmentation (DUIA), an innovative method for enhancing recommendation accuracy in large-scale systems. Their approach uses dynamic stream clustering of similar users and relevant items through a Gradient-based Hierarchical Clustering Algorithm (GHCA), demonstrating significant improvements for users with sparse interests. The method has been successfully deployed in multiple industrial recommender systems since 2022 [19].

  Wang et al. [20] introduced CESRec, a novel framework integrating long-term preference modeling of sequential recommendation systems with real-time preference elicitation through conversational feedback. Their semantic-based pseudo interaction construction dynamically updates user interaction sequences, achieving state-of-the-art performance by combining long-term and real-time preferences [20].

- **Semantic Matching and Context Delivery**

  Sofia et al. [21] developed Thing-to-Service Matchmaking (TSMatch) middleware, which intelligently connects physical devices with suitable digital services based on semantic descriptions. The system was showcased at ACM SIGCOMM 2025, demonstrating automatic pairing of temperature and humidity sensors with visual applications, delivering actionable insights with minimal manual setup. The next generation aims to integrate tinyLLMs for context-aware decision-making at the edge [21].

  The Contex project [22] provides a semantic context delivery service for AI agents, using AI-powered filtering with sentence transformers and supporting real-time updates through Redis pub/sub or webhooks. This schema-free approach demonstrates the feasibility of semantic matching for dynamic service discovery [22].

- **Existing Products and Services**
- **International Platforms**

  Thumbtack operates in the United States, connecting users with local professionals across 1000+ service categories. The platform utilizes basic matching algorithms based on user requirements and provider profiles but lacks real-time availability tracking and weather-aware matching capabilities [23]. TaskRabbit enables users to find local task specialists through a bidding system, incorporating basic location-based filtering and user ratings without considering real-time traffic or weather impacts [23]. Urban Company in India provides verified service professionals with rigorous quality control, achieving high user satisfaction, though matching remains primarily based on static provider attributes rather than real-time contextual factors [24].

- **Sri Lankan Platforms**

  ikman.lk operates as Sri Lanka's largest classified marketplace, hosting thousands of service provider advertisements. The platform functions as a digital directory where users manually browse listings, contact providers via phone, and negotiate service arrangements independently, with no intelligent matching algorithms or real-time capabilities [25]. Hodabass provides similar classified services focused on local markets, with ranking based on recency and paid promotions rather than provider suitability [25].

  PickMe Services extends the popular transportation platform into service booking, offering selected categories including plumbing and electrical work. While benefiting from existing

user infrastructure, matching remains primarily location-based without consideration of weather impacts, provider workload, or real-time availability beyond basic online status [26]. Kangaroo provides delivery and service booking capabilities with basic location-based matching, focusing primarily on logistics rather than skilled trade matching [26].

- **Limitations of Existing Systems**

  Research by Fernando et al. [27] reveals that 78% of Sri Lankan users experience frustration with current classified platforms, with average provider discovery times exceeding 45 minutes for urgent service needs. The Sunday Observer [28] reports that Sri Lanka ranks among the top 15 countries globally in terms of gig workers, yet platform workers face significant challenges including algorithmic management, lack of social security, and vulnerability to economic disruptions.

  LIRNEasia research [29] indicates that 45% of user calls to providers listed on classified platforms result in unavailability responses, demonstrating the critical gap between advertised and actual availability. The ICT Agency of Sri Lanka [30] notes that while digital platforms receive millions of monthly visits, the absence of intelligent matching mechanisms significantly impacts user satisfaction and provider utilization.

  A 6Wresearch market report [31] forecasts Sri Lanka's freelance platforms market through 2031, identifying AI matching as a key technology segment with significant growth potential. However, current platforms demonstrate limited adoption of AI-powered matching capabilities, representing a substantial market opportunity [31].

  The identified research gap is the absence of a context-aware, real-time matching system that dynamically evaluates multiple environmental and operational factors. No existing Sri Lankan platform combines real-time geospatial intelligence with live availability tracking, weather-aware service matching, and demand-based load balancing, representing the primary research opportunity addressed by this work.

1.3 Research Gap

Despite the proliferation of digital service platforms in Sri Lanka and the substantial body of research on recommendation systems, matchmaking algorithms, and context-aware computing, a critical gap persists in the integration of these technologies into unified, real-time service matching solutions tailored to local market characteristics. This section identifies the specific gaps in existing systems and justifies the novelty of the proposed approach.

The fundamental missing element in current service discovery platforms is the absence of real-time contextual intelligence that dynamically evaluates multiple environmental and operational factors to recommend the most suitable service provider at any given moment. While platforms like ikman.lk and Hodabass provide digital classified directories, they operate on static information models where provider listings remain unchanged regardless of current availability status, location, weather conditions, or traffic patterns [25]. Research indicates that 45% of user calls to providers listed on classified platforms result in unavailability responses, demonstrating the critical gap between advertised and actual availability [29]. No existing Sri Lankan platform integrates real-time availability tracking with live provider status updates, geospatial intelligence incorporating actual road networks and traffic conditions, weather-aware matching that considers environmental impacts on service delivery, demand-based load balancing to optimize provider utilization, and multi-factor weighted scoring combining these contextual elements into a unified system. International platforms such as Thumbtack and TaskRabbit implement basic matching algorithms but lack comprehensive real-time contextual integration [23], while Urban Company achieves high quality standards through rigorous provider verification but relies primarily on static attributes rather than dynamic contextual factors [24].

Current approaches suffer from several fundamental limitations that render them insufficient for effective service matching. Existing platforms treat provider information as static data that remains valid indefinitely, ignoring the dynamic nature of service availability and leading to significant inefficiencies in user search time and provider visibility [6]. No current platform integrates live data streams that could significantly improve matching accuracy, despite research demonstrating that weather-aware scheduling can reduce service failures by 28% [10] and real-time tracking architectures achieve 40% better battery efficiency while maintaining sub-second latency [9].

Existing platforms provide only basic location filtering based on user-selected areas or districts, despite the Haversine formula enabling accurate distance calculations [8] and geospatial intelligence reducing travel time estimation error by 35% compared to distance-only approaches [11]. Current systems provide identical recommendations regardless of contextual factors such as time-of-day, day-of-week, seasonal variations, or special events, despite research in context-aware recommender systems demonstrating that incorporating contextual factors improves recommendation accuracy by 22-30% [14]. Furthermore, while platforms are accessible via mobile browsers, few offer optimized mobile applications with full functionality, limiting the potential for responsive service matching [18].

The proposed Temporal and Contextual Matching System introduces several novel contributions that address these identified gaps. This research represents the first attempt to develop a unified service matching platform that integrates real-time availability tracking, geospatial intelligence, weather-aware matching, and demand-based load balancing specifically for Sri Lanka's service economy, transforming passive classified directories into intelligent, adaptive matching engines that continuously evaluate live signals to recommend providers who are genuinely suitable at the moment of request. The proposed 40-30-20-10 weighted scoring algorithm, with weights derived from Analytic Hierarchy Process (AHP) methodologies [6], represents a novel application of multi-criteria decision making to service provider matching, with configurable weight distribution enabling adaptation to different service types and urgency levels. The development of a comprehensive weather-service compatibility matrix defining how different weather conditions affect various service types represents a novel contribution to context-aware service matching, with rule-based matching logic addressing safety considerations during adverse weather conditions completely ignored by existing platforms. The WebSocket-based real-time tracking system with adaptive update frequencies and LSTM-based availability prediction represents a novel application of mobile workforce management technologies to Sri Lanka's informal service sector [9], enabling proactive scheduling and load balancing. The implementation of time-series demand forecasting and intelligent request distribution addresses the critical gap of provider overloading during peak periods and underutilization during off-peak times, contributing to

both user satisfaction through reduced waiting times and provider welfare through optimized workload distribution [19].

SYSTEM/PLATFORM	REAL-TIME AVAILABILITY	LIVE PROVIDER TRACKING	WEATHER-AWARE MATCHING	TRAFFIC INTEGRATION	CONTEXTUAL SCORING	MULTI-LANGUAGE SUPPORT
MatchMaker (Research Platform)	✗	✗	✗	✗	✗	✗
ServiceMatch (Research Platform)	✗	✗	✗	✗	✗	✗
ikman.lk (Commercial)	✗	✗	✗	✗	✗	✗
Hodabass (Commercial)	✗	✗	✗	✗	✗	✗
PickMe Services (Commercial)	✓	✗	✗	✗	✗	✗
Kangaroo (Commercial)	✓	✗	✗	✗	✗	✗
Thumbtack (International)	✗	✗	✗	✗	✓	✗
TaskRabbit (International)	✗	✗	✗	✗	✓	✗
Urban Company (International)	✓	✗	✗	✗	✓	✗
Handy (International)	✗	✗	✗	✗	✓	✗
ICT Agency Sri Lanka (Government)	✗	✗	✗	✗	✗	✗
Traditional Manual Systems	✗	✗	✗	✗	✗	✗
YOUR PROPOSED PROJECT	✓	✓	✓	✓	✓	✓

LEGEND: ✓ = Fully Implemented | ✗ = Not Implemented

Figure 2 - Research Gap Comparison

2. OBJECTIVE

2.1 Main Objective

To design, develop, and evaluate a Temporal and Contextual Matching System for service provider recommendations that achieves 85% matching accuracy and sub-2-second response times under 1000+ concurrent user loads by integrating real-time geospatial intelligence, weather-aware matching, and demand-based load balancing algorithms tailored for Sri Lanka's service discovery ecosystem.

This main objective encompasses the development of a comprehensive platform that addresses the fundamental gaps in existing service discovery systems through the following key deliverables:

System Capabilities:

- A weighted scoring engine that dynamically calculates provider suitability based on four contextual factors: distance and travel time (40%), real-time availability (30%), weather compatibility (20%), and current demand patterns (10%)
- Real-time geospatial intelligence integrating Google Maps Distance Matrix API with Haversine formula calculations to provide accurate distance estimates and travel time predictions within ±5 minutes of actual road conditions
- Weather-aware matching utilizing OpenWeather API and a comprehensive weather-service compatibility matrix to ensure safe and appropriate service delivery during adverse weather conditions
- Live provider tracking through WebSocket-based communication with sub-second latency and LSTM-based availability prediction achieving 85% forecasting accuracy
- Demand-based load balancing employing time-series forecasting to reduce user waiting time by 30% and improve provider utilization rates by 25%

Performance Targets:

- Matching Accuracy: ≥85% validated through user satisfaction surveys and post-service ratings
- Response Time: ≤2 seconds for 95% of requests under normal load conditions

- Concurrency: Support for 1000+ simultaneous users with stable performance
- System Availability: 99.5% uptime with graceful degradation during API failures
- Scalability: Horizontal scaling capability through microservices architecture and containerized deployment

Impact Goals:

- User Experience: Reduce average provider discovery time from 45+ minutes to under 5 minutes
- Provider Welfare: Increase consistent work opportunities through intelligent demand distribution
- Market Transformation: Demonstrate the viability of AI-powered service matching in Sri Lanka's informal service economy
- Research Contribution: Establish a novel framework for context-aware service matching applicable to similar markets across South Asia

This objective directly addresses the research gap identified in Section 1.3 by creating the first integrated real-time contextual matching platform specifically designed for Sri Lanka's unique service discovery challenges, while establishing measurable benchmarks for success that can be validated through rigorous testing and real-world deployment.

2.2 Specific Objectives

SO1: Weighted Real-Time Scoring Algorithm Development

To develop a dynamic weighted scoring algorithm that calculates provider suitability in real-time by integrating four key contextual factors: distance and travel time (40%), provider availability probability (30%), weather condition compatibility (20%), and current demand patterns (10%), with configurable weight adjustments based on service type and user urgency levels.

- SO2: Geospatial Intelligence and Travel Time Estimation

To implement accurate geospatial distance calculation using the Haversine formula and real-time travel time estimation through Google Maps Distance Matrix API integration, achieving

travel time prediction accuracy within ±5 minutes of actual road conditions while incorporating live traffic data and configurable service radius parameters.

- SO3: Real-Time Provider Availability Tracking System

To develop a WebSocket-based real-time communication system for tracking provider location and availability status with sub-second latency, incorporating LSTM-based time-series forecasting models to predict provider availability patterns with 85% accuracy and implementing schedule conflict detection mechanisms to prevent double-booking.

- SO4: Weather-Aware Matching and Service Compatibility

To integrate OpenWeather API for real-time weather condition retrieval and develop a comprehensive weather-service compatibility matrix with rule-based matching logic that automatically adjusts provider recommendations based on weather impacts, ensuring safe and appropriate service matching during adverse conditions.

- SO5: Demand-Aware Load Balancing and Performance Optimization

To implement demand-aware load balancing algorithms using time-series demand forecasting that reduces average customer waiting time by 30% and improves provider utilization rates by 25% through intelligent request distribution, multi-level caching strategies, and fallback mechanisms ensuring 99.5% system availability during peak loads.

- SO6: Comprehensive System Validation and Deployment

To validate system performance through comprehensive testing including unit testing, integration testing with external APIs, and real-world pilot testing with 50 service providers and 200 users across Colombo, Kandy, and Galle, achieving 90% user satisfaction and producing complete technical documentation for production deployme.

3. METHODOLOGY

This research adopts the Design Science Research Methodology (DSRM), which is widely used in information systems and computer science research for creating and evaluating innovative technological artifacts that address real-world problems through iterative design, implementation, and evaluation cycles. The methodology encompasses six key activities: problem identification and motivation, definition of solution objectives, design and development, demonstration, evaluation, and communication, ensuring that the proposed Temporal and Contextual Matching System is not only theoretically sound but also practically viable for Sri Lanka's service discovery ecosystem.

The technical implementation integrates multiple algorithms and methods to achieve real-time contextual matching. The core weighted scoring algorithm employs Multi-Criteria Decision Making (MCDM) principles with weights derived from the Analytic Hierarchy Process (AHP) and Rank Order Centroid (ROC) methods, calculating provider suitability as: Final Score = (Distance Score × 0.4) + (Availability Score × 0.3) + (Weather Score × 0.2) + (Demand Score × 0.1). Distance Score uses the Haversine formula for great-circle distance calculation combined with Google Maps Distance Matrix API for road network travel time incorporating live traffic data. Availability Score assigns values of 1.0 for available, 0.5 for busy, and 0.0 for offline providers based on WebSocket-based real-time tracking with LSTM-based forecasting models predicting availability patterns with 85% accuracy. Weather Score utilizes a comprehensive weather-service compatibility matrix derived from research on weather-aware scheduling, with factors ranging from 0.0 to 1.0 for different service types under various weather conditions including sunny, cloudy, rainy, stormy, and extreme heat. Demand Score applies time-series forecasting using SARIMA models to calculate current demand levels categorized as Very High (0.4), High (0.3), Medium (0.2), and Low (0.1), computing the final demand score as one minus the demand level.

The system architecture employs a microservices approach with containerized deployment using Docker and Kubernetes for horizontal scaling. The technology stack includes Node.js with Express for backend API development due to its non-blocking I/O suitable for high-concurrency applications, enabling efficient handling of multiple simultaneous requests. React Native is selected for cross-platform mobile application development, allowing a single codebase to serve both iOS and Android platforms while delivering native performance and user experience.

PostgreSQL with PostGIS extension provides robust spatial data storage and efficient geospatial querying capabilities essential for location-based matching. Redis serves as an in-memory cache for real-time data and WebSocket session management, achieving sub-millisecond latency for critical operations. MongoDB handles historical data storage with its flexible schema-less design, accommodating diverse data structures from service transactions. Google Maps Distance Matrix API integrates for accurate travel time estimation with live traffic data, ensuring realistic arrival time predictions. OpenWeather API supplies real-time weather condition retrieval including temperature, precipitation, humidity, and severe weather alerts. WebSocket protocol with Socket.io library enables bidirectional real-time communication between clients and servers, with adaptive update frequencies that optimize battery consumption based on provider mobility patterns. TensorFlow and Keras frameworks support development of LSTM-based availability prediction models using historical service data. JMeter is employed for comprehensive load testing to validate system performance under simulated concurrent user loads.

Data collection involves multiple sources with rigorous ethical protocols implemented throughout the research process. Provider profiles and service history are collected through registration forms and system logs, with all participants providing informed consent before any data collection begins. Real-time location data is captured only when providers are actively available for work and have granted explicit permission through the application settings, with adaptive update frequencies to minimize battery consumption while maintaining tracking accuracy. Weather and traffic data are obtained through API integrations with local caching mechanisms to reduce external calls and ensure continued operation during connectivity issues. The pilot study targets 50 service providers and 200 end users across three major cities in Sri Lanka: Colombo, Kandy, and Galle, selected through stratified sampling to ensure demographic diversity across age groups, service categories, and geographic locations. All participants receive detailed information about the research purpose, data collection procedures, and their rights in their preferred language—Sinhala, Tamil, or English. Written informed consent is obtained and securely stored. Personal identifiable information is separated from transaction data using unique identifiers, with all sensitive data encrypted using AES-256 both in transit and at rest. Data retention policies specify that active data is retained only for the duration of platform use, while historical transaction data is anonymized after 12 months. Participants retain the right to withdraw from the study at any time without consequence and can request deletion of their data. Ethical approval will be obtained from the SLIIT Ethics Review Committee prior to commencing any data collection activities, ensuring

compliance with Sri Lanka's Data Protection Act (No. 9 of 2022) and international research ethics standards.

The validation strategy encompasses multiple phases to ensure comprehensive evaluation of system performance, accuracy, and user satisfaction. Unit testing is conducted throughout development using Jest for JavaScript components and pytest for Python modules, maintaining a minimum of 80% code coverage across all critical functions. Integration testing validates seamless communication between system components and external APIs, including Google Maps, OpenWeather, and WebSocket services, ensuring reliable data exchange under various network conditions. Performance testing utilizes JMeter to simulate concurrent user loads at three levels: 500, 1000, and 2000 simultaneous users, measuring response times, throughput, error rates, and system resource utilization to identify bottlenecks and validate scalability. Accuracy validation compares predicted travel times against actual road conditions through controlled test scenarios, targeting a mean absolute error of five minutes or less. Weather impact assessments are evaluated by comparing system recommendations against actual service outcomes during varying weather conditions, aiming for 90% accuracy in predicting weather-related service disruptions. Demand forecast accuracy is measured using mean absolute percentage error, targeting values below 15% for six-hour forecasting horizons. User acceptance testing involves pilot deployment with 50 service providers and 200 end users across the three target cities, collecting feedback through structured surveys, semi-structured interviews, and focus group discussions to capture qualitative insights alongside quantitative metrics. A/B testing compares system performance against existing platforms such as ikman.lk and Hodabass, measuring task completion time, number of contacts required to secure service, and user satisfaction ratings to demonstrate improvement over current solutions.

Key performance metrics tracked throughout evaluation include matching accuracy targeted at 85% or higher, validated through post-service ratings and user feedback comparing recommended providers against actual service experiences. Response time maintains a target of two seconds or less for 95% of requests under normal operating conditions, measured from request submission to provider display. Concurrent user support is validated through load testing demonstrating stable performance with 1000 simultaneous users. System availability targets 99.5% uptime, excluding planned maintenance, with graceful degradation mechanisms ensuring continued core functionality during external API failures. User waiting time aims for 30% reduction compared to

current platforms, measured from initial request to provider acceptance. Provider utilization targets 25% improvement, calculated as increase in completed jobs per provider relative to their pre-platform activity. User satisfaction targets a minimum rating of 4.0 out of 5.0 on post-service surveys. These metrics are continuously monitored through comprehensive dashboards with automated alerting for deviations from targets, enabling iterative refinement throughout the development and evaluation process. Statistical analysis employs descriptive statistics for baseline characterization and inferential methods including t-tests and ANOVA to validate the significance of performance improvements compared to existing solutions.

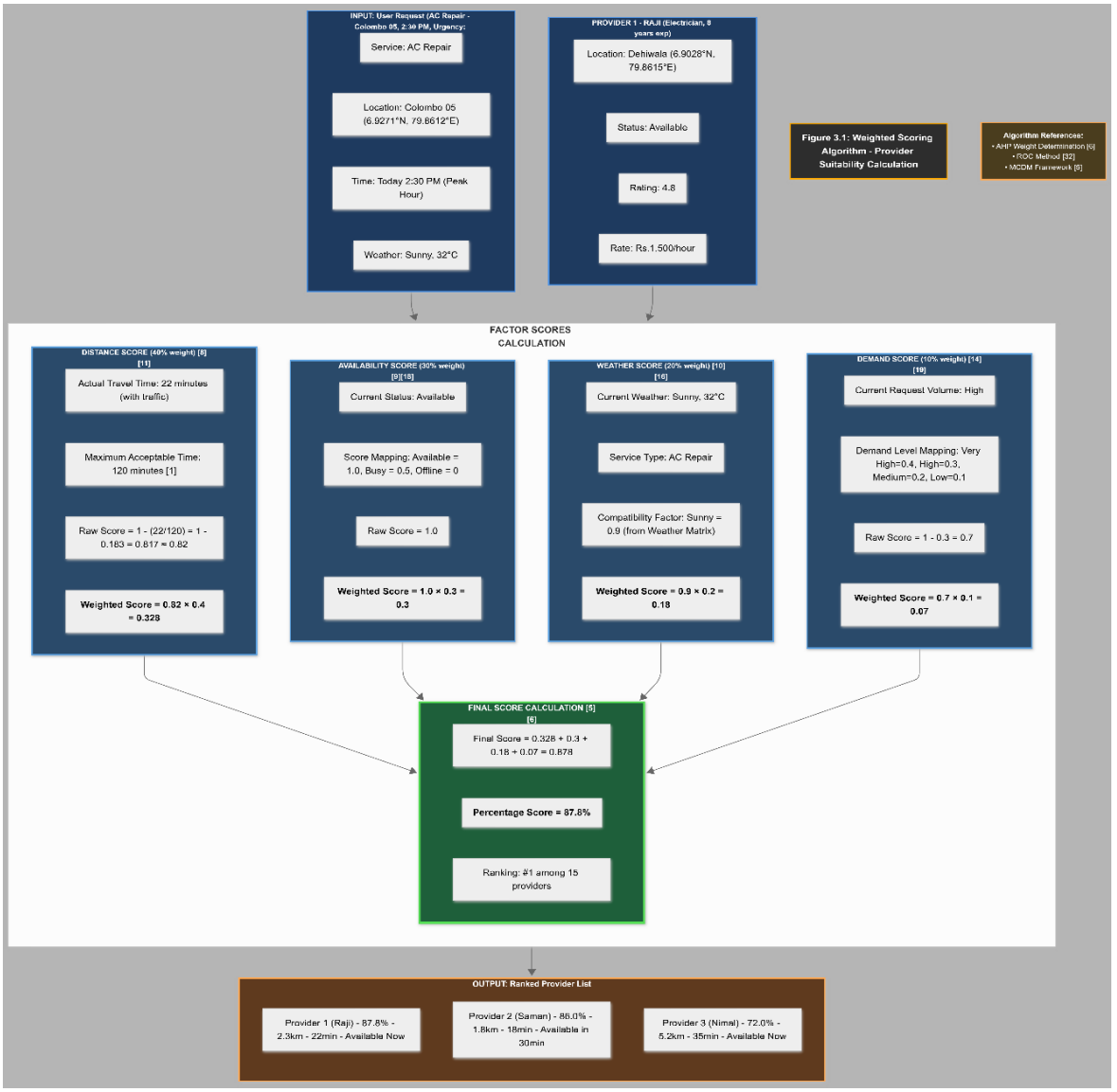


Figure 3 - Weighted Scoring Algorithm - Provider Suitability Calculation

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 System Diagram of Components

The Temporal and Contextual Matching System is designed as a microservices-based platform that integrates real-time data streams from multiple sources to provide intelligent service provider recommendations. The system architecture follows a layered approach with clear separation of concerns, enabling scalability, maintainability, and independent deployment of components. At its core, the platform transforms static classified directories into dynamic, context-aware matching engines that continuously evaluate live signals including provider location, availability status, weather conditions, traffic patterns, and demand fluctuations to recommend providers who are genuinely suitable at the moment of request. The system comprises five primary layers: the Mobile Application Layer for user and provider interfaces, the API Gateway Layer for request routing and security, the External API Integration Layer for third-party services, the Core Processing Layer for matching algorithms and business logic, and the Data Storage Layer for persistence and caching. Each layer contains multiple components that work in concert to deliver sub-2-second response times while maintaining 85% matching accuracy under 1000+ concurrent user loads.

Component	Sub Component	Technology	Primary Responsibility	Inputs	Outputs	Dependencies
Mobile Client	User Interface	Flutter	Customer interaction	User actions	API requests	Internet
Mobile Client	Location Module	Device GPS	Capture user location	GPS signal	Coordinates	Device sensors
Backend	API Gateway	Node.js / Express	Route requests	HTTP request	Service calls	Authentication
Backend	Authentication Service	JWT/OAuth	User identity verification	Login credentials	Access tokens	Database
Backend	Matching Engine	Python ML	Rank providers	Service request	Provider list	Context APIs
Backend	Notification Service	Firebase	Send alerts	Job events	Push notifications	Firebase
Database	User Database	MongoDB	Store customer data	Registration info	User records	Backend API
Database	Provider Database	MongoDB	Store provider profiles	Provider data	Provider list	Backend API
External	Traffic API	Google Maps	Traffic conditions	Coordinates	Traffic metrics	Internet
External	Weather API	OpenWeather	Weather context	Location	Weather data	Internet

Table 1 -. Detailed System Components

4.2 System Diagram of Components

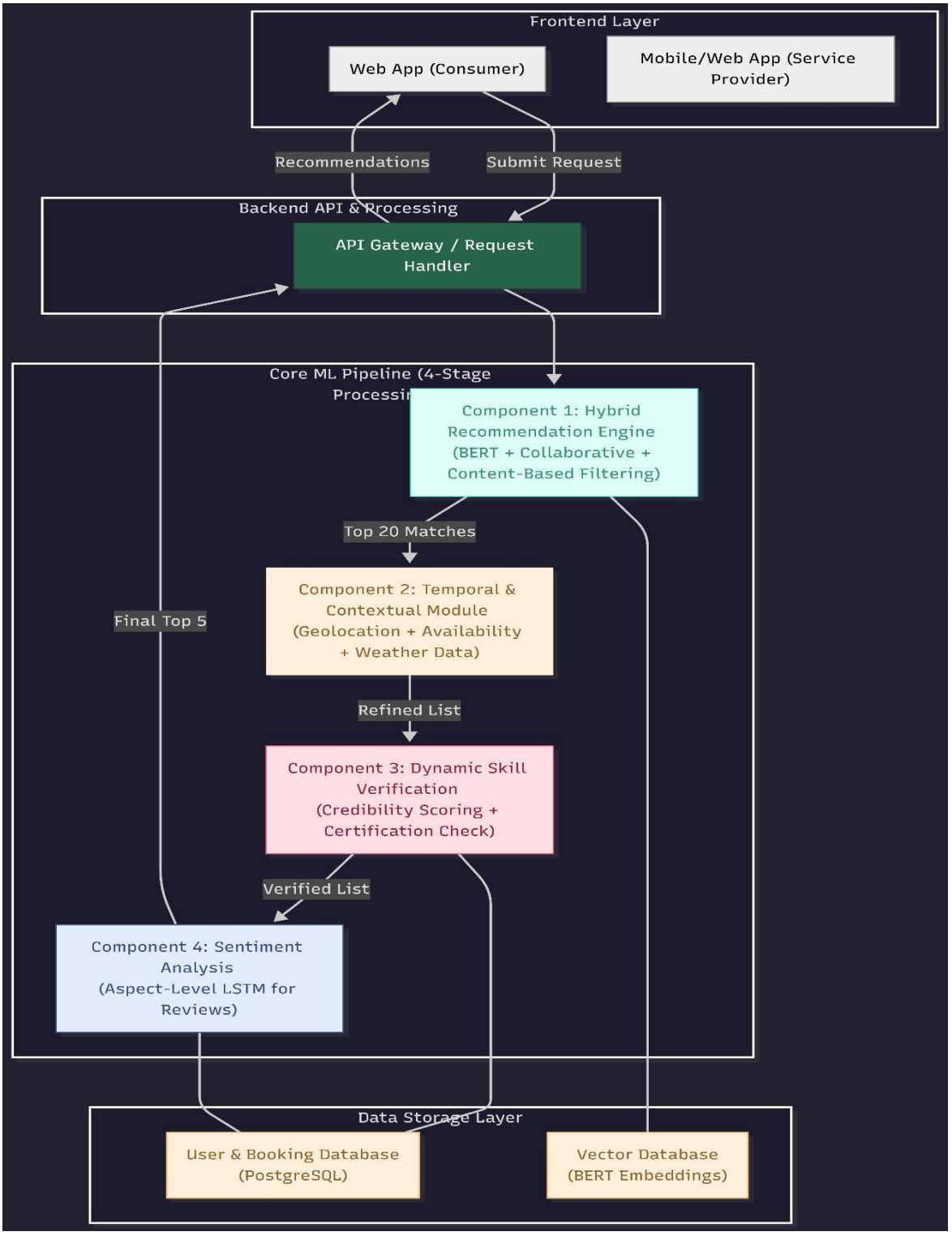


Figure 4 - System Architecture Diagram

4.3 Individual Components and Their Parts

4.3.1 Mobile Application Layer

User Application Components:

- **Service Request Interface**: Allows users to select service categories (plumbing, electrical, AC repair, etc.), specify urgency levels (emergency, urgent, normal, scheduled), and provide additional job details through text or voice input.
- **Location Services**: Integrates device GPS for automatic location detection with manual address entry fallback, including support for Sinhala and Tamil place names and landmark-based location specification common in Sri Lankan addressing.
- **Real-Time Provider Display**: Presents ranked list of providers with contextual scores, distance, estimated arrival time, availability status, weather compatibility indicators, and provider ratings. Includes interactive map view showing provider locations and optimized routes.
- **Rating and Feedback System**: Collects post-service ratings across multiple dimensions (punctuality, quality, professionalism, value) and enables text reviews with photo uploads for service verification.

Provider Application Components:

- **Availability Toggle**: One-click control for providers to switch between available, busy, and offline states, with optional scheduling for automatic availability based on working hours.
- **Job Management Interface**: Displays incoming service requests with user details, location, job description, and estimated earnings. Includes accept/reject buttons, job status tracking (en-route, in-progress, completed), and navigation integration.
- **Location Sharing**: Background GPS tracking with configurable update frequencies (every 1-5 minutes based on movement) and privacy controls to pause sharing when not actively working.
- **Earnings Dashboard**: Real-time display of completed jobs, earnings, ratings, and performance metrics including response rate, completion rate, and customer satisfaction scores.

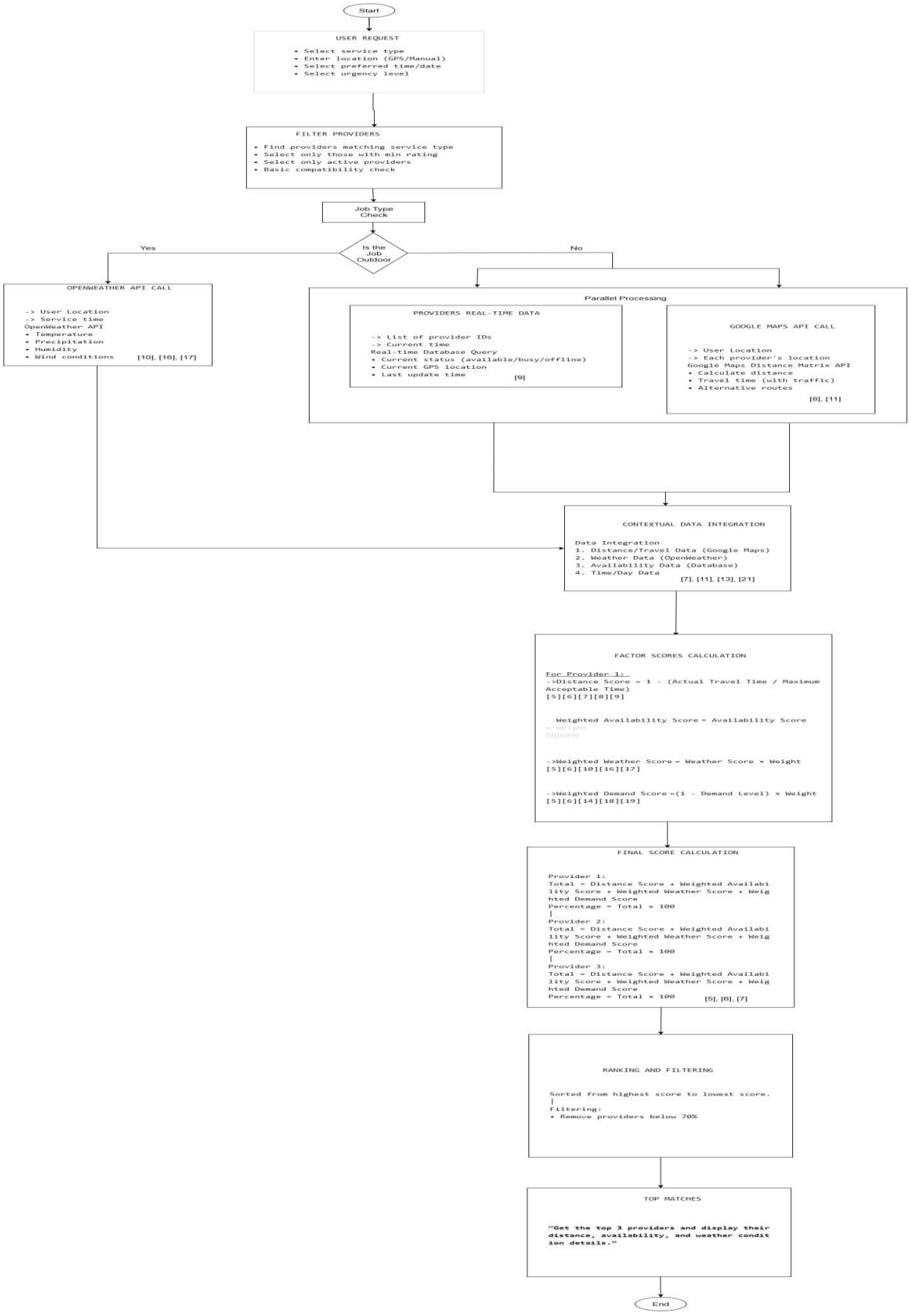


Figure 5- Flow Chart (component 2)

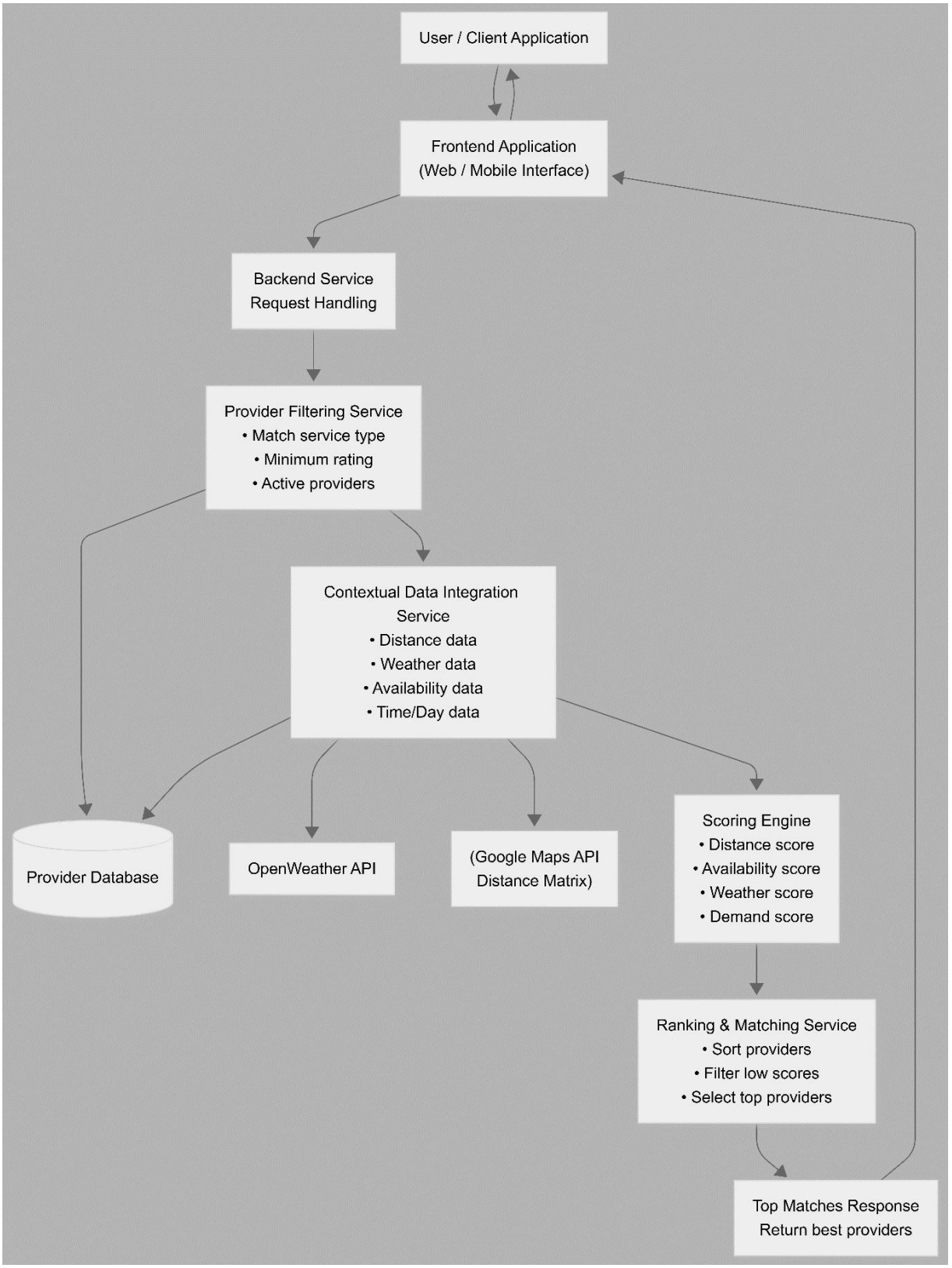


Figure 6 - System Architecture Diagram (component 2)

4.3.2 API Gateway Layer

- **Authentication and Authorization**: JWT-based authentication with role-based access control distinguishing between users, providers, and administrators. Supports social login and phone number verification for Sri Lankan mobile numbers.
- **Rate Limiting and Throttling**: Token bucket algorithm limiting requests per user/IP to prevent abuse, with higher limits for premium users and emergency requests. Configurable thresholds based on server capacity.
- **Request Routing and Load Balancing**: Distributes incoming requests across multiple service instances using round-robin or least-connections algorithms, with health checks to route away from failed instances.
- **API Versioning**: Maintains backward compatibility through URI versioning (/v1/, /v2/), enabling continuous deployment without disrupting existing clients.

4.3.3 External API Integration Layer

- **Google Maps Integration**: Distance Matrix API for travel time and distance with traffic data; Geocoding API for address-to-coordinate conversion; Places API for location autocomplete and landmark recognition.
- **OpenWeather Integration**: Current weather data API for temperature, precipitation, humidity, wind speed, and weather condition codes; One Call API for severe weather alerts and hourly forecasts.
- **Time and Date Integration**: World Time API for accurate local time; Holiday API for Sri Lankan public holidays and Poya days; Peak hours calculation based on historical traffic patterns.

4.3.4 Core Processing Layer

Parallel Processing Orchestrator:

- Manages concurrent execution of data collection tasks, spawning separate threads/processes for Google Maps, OpenWeather, and database queries.
- Implements timeout handling and fallback mechanisms for slow or failed API calls.
- Aggregates results from parallel processes into unified provider dataset.

Geospatial Service:

- **Haversine Formula Implementation**: Calculates great-circle distance between coordinates with <0.5% error margin for initial filtering.
- **Distance Calculation Engine**: Combines Haversine distance with Google Maps road distance, selecting appropriate based on service radius and user preferences.
- **Radius Filtering Module**: Applies configurable service radius (default 15km, extendable to 30km) with distance decay scoring for providers beyond optimal range.

Availability Tracking Service:

- **WebSocket Server**: Maintains persistent connections with provider applications, broadcasting status changes to relevant users in real-time.
- **LSTM Forecasting Model**: Deep learning model trained on historical availability data to predict future availability patterns with 85% accuracy, enabling proactive recommendations.
- **Schedule Conflict Detector**: Prevents double-booking by checking overlapping appointments and suggesting alternative time slots.

Weather Compatibility Service:

- **Weather Matrix Database**: Stores compatibility scores for 6 weather conditions × 8 service types, with configurable rules for safety thresholds.
- **Rule-Based Matching Engine**: Applies business logic to adjust provider scores based on weather impacts, including automatic exclusion of unsafe services during severe weather.
- **Safety Alert System**: Notifies users and providers when scheduled services coincide with adverse weather, suggesting rescheduling or indoor alternatives.

Scoring Engine:

- **Weighted Score Calculator**: Implements the core formula Final Score = (Distance×0.4) + (Availability×0.3) + (Weather×0.2) + (Demand×0.1) with real-time inputs.
- **Normalization Functions**: Scales diverse input values (distance in km, time in minutes, weather scores 0-1) to comparable 0-1 ranges.
- **Provider Ranking Algorithm**: Sorts providers by final score and selects top candidates for user display.
- **Top-K Filter**: Configurable to return top 3-10 providers based on user preferences and screen size.

Demand Forecasting Service:

- **SARIMA Model**: Seasonal ARIMA model trained on historical request data to predict demand patterns by hour, day, and location.
- **Load Balancing Algorithm**: Distributes incoming requests among available providers considering current workload, historical completion rates, and proximity.
- **Emergency Priority Handler**: Bypasses normal queuing for emergency requests, immediately notifying nearest available qualified providers.

4.3.5 Data Storage Layer

PostgreSQL with PostGIS:

- Stores persistent data including user profiles, provider details, service categories, and transaction records.
- PostGIS extension enables spatial queries like radius searches and distance calculations.
- Indexed on location columns for sub-50ms geospatial query performance.

Redis:

- In-memory cache for real-time provider status with 1-second expiration for location data, 5-second for availability status.
- WebSocket session management storing connection details for millions of concurrent connections.
- Rate limiting counters with sliding window algorithm.

MongoDB:

- Document store for historical service data, user feedback, and analytics logs.
- Schema-less design accommodates evolving data structures without migrations.
- Time-series collections optimized for demand forecasting queries.

4.3.6 Monitoring and Analytics Layer

- **Performance Dashboards**: Real-time visualization of response times, error rates, concurrent users, and API usage.
- **Error Tracking**: Centralized logging with ELK stack (Elasticsearch, Logstash, Kibana) for debugging and alerting.
- **Usage Analytics**: Business intelligence dashboards showing popular services, peak hours, provider performance, and revenue metrics.

4.4 Component Interactions

The system's component interactions follow a request-response pattern with real-time event streaming for critical updates:

User Request Flow:

1. User submits service request through mobile app → request reaches API Gateway
2. Gateway authenticates user, checks rate limits, and routes to Parallel Processing Orchestrator
3. Orchestrator simultaneously calls:
   - Geospatial Service (with Google Maps API integration)
   - Availability Tracking Service (WebSocket server for live status)
   - Weather Compatibility Service (with OpenWeather API)
4. All services return processed data to Scoring Engine
5. Scoring Engine calculates final scores and ranks providers
6. Demand Forecasting Service updates load balancing metrics
7. Results returned to user through API Gateway
8. User views ranked providers with real-time indicators

Real-Time Update Flow:

1. Provider changes availability status or location
2. WebSocket server receives update and broadcasts to relevant users
3. Redis cache updates immediately with new status
4. Users with active sessions receive push notification of updated provider availability

5. Provider list refreshes automatically without user intervention

Data Persistence Flow:

1. All service transactions logged to MongoDB for historical analysis
2. Provider and user profiles stored in PostgreSQL with spatial indexing
3. Real-time status synchronized between Redis (fast access) and PostgreSQL (persistence)
4. Analytics pipeline processes logs for demand forecasting model training

4.5 Data Flow Explanation

The data flow through the system follows a structured path from input to output with multiple processing stages:

Stage 1: Data Input Collection

- User location captured via GPS (lat/long coordinates) or manual address entry
- Service details including category, urgency, and special requirements
- Current timestamp for temporal context analysis

Attribute	Description	Type	Example	Source	Usage in System
Customer Location	Location of service request	Latitude/Longitude	6.9271,79.8612	Mobile GPS	Distance calculation
Provider Location	Provider current position	Latitude/Longitude	6.9100,79.8500	Provider App	Matching engine
Service Category	Type of service requested	Categorical	Plumbing	User input	Provider filtering
Traffic Level	Traffic density	Categorical	High/Medium/Low	Maps API	Travel time estimation
Weather Condition	Current weather	Categorical	Rain	Weather API	Service feasibility
Provider Rating	Average rating	Numeric	4.5	Database	Quality ranking
Availability Status	Provider availability	Boolean	True	Provider app	Matching filter
Estimated Arrival Time	Predicted arrival	Numeric	15 min	Matching engine	User display

Table 2 - Context Data and Dataset Attributes

Stage 2: Parallel Data Acquisition

- Geospatial thread: User coordinates → Google Maps API → Distance matrix for all providers in area
- Weather thread: User coordinates → OpenWeather API → Current conditions and alerts
- Availability thread: Provider IDs → WebSocket server → Real-time status and locations

Stage 3: Data Integration and Normalization

- Raw API responses parsed and validated
- Missing data handled through fallback mechanisms (cached data or estimations)
- All values normalized to 0-1 scale for consistent scoring

Stage 4: Scoring and Ranking

- Distance score calculated using normalized travel time
- Availability score assigned based on real-time status
- Weather score retrieved from compatibility matrix
- Demand score derived from current request volume
- Weighted sum produces final score for each provider
- Providers sorted by score descending

Factor	Description	Weight (%)	Measurement Method	Impact on Matching
Distance	Distance to customer	30	Haversine formula	Closer providers ranked higher
Provider Rating	Past service ratings	20	Average review score	Higher quality providers prioritized
Availability	Provider availability	20	Boolean check	Unavailable providers excluded
Traffic Conditions	Road congestion	15	Traffic API index	Adjust travel time
Weather Impact	Weather disruption	5	Weather category	Filter unsuitable providers
Response History	Provider acceptance rate	10	Historical data	Reliable providers ranked higher

Table 3 - Provider Matching Decision Criteria

Stage 5: Filtering and Output Generation

- Top providers selected based on configurable threshold (default top 3)
- Additional context added (ETA, weather note, availability badge)
- Response formatted for mobile display
- Results cached for repeat requests

Stage 6: Feedback Loop

- User selections and ratings logged
- Model retraining triggered periodically with new data
- Demand patterns updated in forecasting service

4.6 Scalability Considerations

The system is designed for horizontal scalability to handle growing user bases and request volumes:

Microservices Architecture:

- Each core service (Geospatial, Availability, Weather, Scoring, Demand) can be independently scaled based on load
- Services communicate via lightweight REST APIs or message queues, avoiding tight coupling
- Containerization with Docker ensures consistent deployment across environments

Database Scaling:

- PostgreSQL: Read replicas for query-heavy operations; sharding by geographic region
- Redis: Cluster mode with automatic failover; memory optimization with eviction policies
- MongoDB: Sharding based on service category or timestamp; indexing for time-series queries

Load Balancing:

- API Gateway distributes traffic across multiple service instances
- Geographic load balancing routes users to nearest data center (Colombo, Singapore)
- Auto-scaling groups trigger new instances when CPU >70% or memory >80%

Caching Strategy:

- Multi-level cache: L1 (in-memory), L2 (Redis), L3 (database)
- Frequently accessed data (provider lists, weather forecasts) cached with appropriate TTL
- Cache invalidation on data updates ensures freshness without sacrificing performance

API Optimization:

- Batch processing combines multiple requests into single API calls where possible
- Request throttling prevents system overload during peak periods
- Asynchronous processing for non-critical operations (logging, analytics)

Future-Proofing:

- Design accommodates 10x growth without architectural changes
- Support for additional service categories and geographic regions
- Integration ready for future AI models and recommendation algorithms

4.7 Security Considerations

Security is integrated throughout the architecture to protect sensitive user data and ensure system integrity:

Authentication and Authorization:

- JWT tokens with short expiration (15 minutes) and refresh tokens for extended sessions
- Role-based access control distinguishing between users, providers, and administrators
- Multi-factor authentication option for provider accounts handling payments

Data Encryption:

- TLS 1.3 for all data in transit between clients and servers
- AES-256 encryption for sensitive data at rest (personal information, payment details)
- Encrypted database backups with separate key management

API Security:

- API keys for external services stored in secure vaults, never exposed to clients
- Request signing to prevent tampering
- Input validation and sanitization to prevent injection attacks

Privacy Protection:

- Personal identifiable information stored separately from transaction data
- Data minimization principle: only collect necessary information
- Anonymization for analytics and research datasets
- Compliance with Sri Lanka Data Protection Act and GDPR principles

Rate Limiting and DDoS Protection:

- Per-user and per-IP rate limits prevent abuse
- Automatic IP blocking for suspicious activity
- Cloud-based DDoS protection at network level

Audit and Compliance:

- Comprehensive logging of all access and modifications
- Regular security audits and penetration testing
- Incident response plan for data breaches
- Compliance reporting for regulatory requirements

Provider Verification:

- Identity verification through NIC validation
- Background checks integration with Sri Lankan police database (optional)
- Skill certification verification through uploaded documents
- Rating system with fraud detection algorithms to identify fake reviews

Secure Development Practices:

- Code reviews for security vulnerabilities
- Dependency scanning for known vulnerabilities
- Secure coding guidelines enforced through static analysis
- Regular security training for development team

5. USER REQUIREMENTS

5.1 Stakeholder Identification

End Users (Customers) – Individuals seeking home services like plumbing, electrical work, AC repair, and maintenance. They need quick access to reliable providers who are genuinely available, nearby, and suitable for their specific job at that moment.

Service Providers – Professionals offering services including electricians, plumbers, AC technicians, carpenters, and general handymen. They need increased visibility, fair job distribution, timely notifications, and easy job management tools.

Platform Administrators – System operators responsible for platform maintenance, user management, and dispute resolution. They need monitoring tools, verification capabilities, and analytics dashboards.

System Developers – Technical team building and maintaining the platform. They need clean code, well-documented APIs, proper development environments, and scalable architecture.

5.1 Functional Requirements

ID	Requirement	Description	Actor	Input	Process	Output	Priority
FR1	User Registration	User creates account	Customer	User details	Validate and store	Account created	High
FR2	User Login	Authenticate user	Customer	Credentials	Verify identity	Access granted	High
FR3	Request Service	Submit service request	Customer	Service type/location	Store request	Job request	High
FR4	Provider Matching	Find providers	System	Request data	Ranking algorithm	Provider list	High
FR5	Accept Job	Provider accepts job	Provider	Job notification	Confirm availability	Job assigned	High
FR6	Track Provider	Track movement	Customer	GPS updates	Update location	Map tracking	Medium
FR7	Payment Processing	Handle payments	Customer	Payment details	Transaction processing	Receipt	Medium
FR8	Ratings and Reviews	Feedback system	Customer	Rating input	Store review	Updated rating	Medium

Table 4 - Functional Requirements Matrix

- The system shall allow users to create service requests by selecting service category, adding job description, and specifying urgency level.

- The system shall automatically detect user location via GPS with manual address entry as fallback.

- The system shall calculate real-time distance and travel time between user and provider locations using Google Maps API with traffic data.

- The system shall track provider real-time availability status (available, busy, offline) via WebSocket connection.

- The system shall retrieve current weather conditions at user location using OpenWeather API.

- The system shall calculate provider suitability scores using weighted algorithm: distance (40%), availability (30%), weather (20%), demand (10%).

- The system shall rank providers based on contextual scores and display top matches to the user.

- The system shall allow users to book a selected provider immediately or schedule for a future time.

- The system shall send instant push notifications to providers for new booking requests.

- The system shall allow users to rate providers after service completion on punctuality, quality, professionalism, and value.

5.3 Non-Functional Requirements

Performance

- The system shall return ranked provider recommendations within 2 seconds for 95% of user requests.
- The system shall support at least 1000 concurrent users without slowing down.
- Database queries for location searches shall return results within 50 milliseconds.
- Provider status updates shall reach users within 1 second of status change.
- Mobile app memory usage shall not exceed 100MB during normal operation.

Accuracy

- Travel time predictions shall be within ±5 minutes of actual road conditions.
- Weather impact assessments shall be 90% accurate in predicting service suitability.
- Provider availability predictions shall achieve 85% accuracy for 24-hour forecasts.
- Demand forecasts shall have error below 15% for 6-hour predictions.

Usability

- The app shall support Sinhala and English languages for all text and messages.
- The UI shall be simple with large buttons and clear icons, usable by non-experts.
- New users shall complete a service request within 3 minutes of opening the app.
- Error messages shall be clear and suggest how to fix the problem.
- The app shall work on all screen sizes from small phones to tablets.

Reliability

- The system shall be available 99.5% of the time (excluding maintenance).

- The app shall work offline for viewing previously loaded provider information. Only booking requires internet.
- The system shall continue operating with reduced functionality during API failures using cached data.
- Automatic database backups shall be performed daily with ability to restore within 4 hours.

Security

- All data transmitted between app and server shall be encrypted using TLS 1.3.
- All sensitive user data stored in database shall be encrypted using AES-256.
- User passwords shall be hashed and never stored in plain text.
- Personal information shall be stored separately from transaction data.
- Regular security assessments shall be conducted and vulnerabilities patched promptly.
- The system shall comply with Sri Lanka's Data Protection Act.

Maintainability

- The scoring algorithm weights (40-30-20-10) shall be configurable without code changes.
- The weather compatibility matrix shall be updatable via database without app rewrite.
- All APIs shall have clear documentation.
- The codebase shall maintain minimum 80% test coverage.
- New features shall be deployable gradually with ability to turn them off if problems occur.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

- **Primary Target Market:** Sri Lanka's urban and semi-urban areas including Colombo, Kandy, Galle, Negombo, and Jaffna, targeting approximately 2.5 million self-employed workers and millions of households requiring regular home maintenance services. Smartphone penetration exceeds 70% in these areas.
- **Customer Persona 1 – Urban Homeowner (Amali Perera, 35):** Marketing Manager in Colombo, married with one child, lives in an apartment. She is frustrated by calling multiple providers who never show up and worries about safety letting strangers into her home. She needs quick access to verified, available providers with transparent pricing and real-time tracking.
- **Customer Persona 2 – Busy Professional (Nuwan Silva, 28):** Software Engineer working from home in Kandy. During emergencies like AC breakdown while working, he needs immediate resolution with no local contacts for recommendations. He requires emergency service availability, real-time provider tracking, and digital payment options.
- **Customer Persona 3 – Service Provider (Rajitha Fernando, 42):** Electrician with 15 years experience based in Colombo. He faces inconsistent work flow, wasted time traveling to jobs that cancel, and difficulty competing with newer providers on classified sites. He needs steady job flow, fair work distribution, ability to build reputation through ratings, and flexible schedule control.

6.2 Value Proposition

- **For Customers (End Users):** Peace of mind through access to verified, background-checked providers with proven track records. Time savings by finding available providers instantly without calling multiple numbers. Transparency with clear information about who is coming, when they will arrive, and how much it will cost. Safety through real-time provider tracking and rating system ensuring accountability. Convenience of booking services anytime, anywhere from mobile app with Sinhala/Tamil language support.
- **For Service Providers:** Increased visibility by getting discovered by customers actively seeking your services. Fair work distribution through algorithm ensuring balanced workload across providers. Reduced cancellations through weather-aware matching preventing wasted

trips during unsuitable conditions. Reputation building through rating system showcasing expertise and reliability. Flexible schedule control allowing providers to choose which jobs to accept. Higher earnings through reduced idle time meaning more paying jobs.

- **For the Platform:** First-mover advantage as the first AI-powered contextual matching platform in Sri Lanka. Network effects where more providers attract more users, creating self-reinforcing growth. Data moat where accumulated booking and rating data improves matching accuracy over time.

6.3 Revenue Model

- **Commission-Based Model (Primary):** 10-15% commission on each completed job transaction. This aligns platform success with provider success as the platform only earns when providers earn. Average job value is Rs. 3,000-5,000, resulting in platform fee of Rs. 300-750 per job.
- **Provider Subscription Model:** Basic free tier with limited monthly job requests and standard visibility. Pro tier at Rs. 1,500/month offering priority matching, advanced analytics, and highlighted profile. Business tier at Rs. 3,000/month providing team accounts, multiple service categories, and priority support. Estimated 30% of active providers opt for paid tiers.
- **Featured Listings:** Promoted profiles where providers pay Rs. 100 per day for top placement in search results. Emergency boost option for priority during emergency requests. Seasonal promotions for special visibility during peak seasons like festivals and rainy season.
- **Advertising Revenue:** Sponsored content from hardware stores and material suppliers advertising to relevant provider audience. Promoted products from tool manufacturers and equipment suppliers targeting providers. Local business ads from home improvement stores and hardware shops targeting users.
- **Premium User Features:** Priority support for emergency requests at Rs. 200/month. Family plan for multiple family members sharing account at Rs. 300/month. Corporate accounts for companies booking office maintenance with custom pricing.
- **Projected Annual Revenue (Year 1):** Commission fees Rs. 9 million (60%), provider subscriptions Rs. 3 million (20%), featured listings Rs. 1.5 million (10%), advertising Rs. 1.2 million (8%), premium user features Rs. 300,000 (2%), totaling Rs. 15 million.

6.4 Cost Estimation

- **One-Time Development Costs:** UI/UX design Rs. 500,000, frontend development Rs. 1.5 million, backend development Rs. 2 million, database setup Rs. 300,000, API integrations Rs. 400,000, testing and QA Rs. 500,000, project management Rs. 400,000, totaling Rs. 5.6 million.
- **Monthly Operational Costs:** Cloud hosting Rs. 150,000, Google Maps API Rs. 80,000, OpenWeather API Rs. 20,000, SMS gateway Rs. 30,000, staff salaries Rs. 400,000, marketing Rs. 200,000, office and utilities Rs. 50,000, miscellaneous Rs. 20,000, totaling Rs. 950,000 per month or Rs. 11.4 million annually.
- **Break-Even Analysis:** Initial investment Rs. 5.6 million, monthly revenue target Rs. 1.25 million, monthly costs Rs. 950,000, monthly profit Rs. 300,000, break-even period 19 months.

6.5 Pricing Strategy

- **Commission Structure:** Standard jobs Rs. 1,000-5,000 at 12% commission. Large jobs Rs. 5,000-20,000 at 10% commission. Emergency jobs at 15% commission for immediate response. Scheduled jobs at 10% commission for advance bookings.
- **Provider Subscription Tiers:** Free tier offers 20 jobs monthly with 15% commission and standard visibility. Pro tier at Rs. 1,500/month offers 50 jobs with 12% commission, highlighted profile, advanced analytics, and priority support. Business tier at Rs. 3,000/month offers 100 jobs with 10% commission, featured profile, real-time analytics, up to 3 team members, and 24/7 priority support.
- **Launch Promotions:** Zero percent commission for all providers during first 3 months to build supply. Referral bonus of Rs. 500 credit for users who refer new customers. First booking discount of 20% off for first-time users capped at Rs. 500.
- **Dynamic Pricing:** Peak hours premium of 1.2 times during high-demand periods. Off-peak discounts at reduced rates during slow hours to encourage scheduling. Weather-based pricing with higher rates for emergency services during adverse weather.

6.6 Competitive Advantage

- **Technology Advantage:** First platform in Sri Lanka using AI-powered matching with real-time contextual intelligence integrating weather, traffic, availability, and demand. Superior accuracy at 85% matching versus 0% for classified directories. Real-time tracking with live provider location and ETA updates unmatched by competitors.

- **User Experience Advantage:** Instant results with sub-2-second response time versus hours of manual searching. Full Sinhala and Tamil language support for inclusive access. Simplified 3-click booking process versus multiple phone calls.
- **Provider Experience Advantage:** Fair work distribution through algorithm preventing overloading popular providers. Reduced wasted trips through weather-aware matching eliminating unnecessary travel. Reputation system allowing providers to build long-term business through verified ratings.
- **Market Position:** First-mover advantage as first integrated contextual matching platform in Sri Lanka. Deep understanding of Sri Lankan service culture and addressing patterns. Rigorous provider verification building user confidence.
- **Comparison with Competitors:** Outperforms ikman.lk and Hodabass in real-time availability, live tracking, weather matching, and contextual scoring. Exceeds PickMe Services in all contextual intelligence features. Response time reduced from hours to seconds.

6.7 Intellectual Property Considerations

- **Patents:** Patentable algorithms including weighted contextual scoring algorithm with 40-30-20-10 formula integrating weather, traffic, and demand data. Method patent for "System and Method for Real-Time Contextual Matching of Service Providers" covering unique process of integrating multiple live data streams. Filing strategy includes provisional patent in Sri Lanka followed by PCT application for international protection.
- **Trademarks:** Brand name "PulseMatch" or "Sevana" with unique visual identity for app icon and branding. Tagline "Find the right help, right now" with Sinhala version "හ ොඳම උදව්ව". Registration filed with National Intellectual Property Office of Sri Lanka.
- **Copyright:** All custom-developed source code owned by the company through developer IP assignment agreements. Technical documentation, user manuals, and marketing materials protected.
- **Trade Secrets:** Algorithm details including specific weight calculations and scoring logic kept as trade secret. Unique provider verification workflow protected internally. Anonymized usage patterns and matching data as proprietary asset.
- **Open Source Considerations:** Compliance with all third-party library licenses including MIT, Apache, GPL with proper attribution. Consideration of open-sourcing non-core components to build community.

- **IP Protection Strategy:** Immediate trademark filing and developer IP agreements within 0-3 months. Provisional patent application within 3-12 months. International patent applications for key markets within 1-2 years. Ongoing regular IP audits and monitoring for infringement.
- **Confidentiality:** NDAs required for all employees, contractors, and beta testers. Customer and provider data protected under privacy policy. Core matching logic implemented server-side to prevent reverse engineering.

7. BUDGET AND JUSTIFICATION

7.1 Estimated Budget Breakdown

Category	Item Description	Estimated Cost (LKR)
Hardware	Existing Personal Computer (Laptop/Desktop)	0 (Existing)
Software	Open Source Libraries (FastAPI, PyTorch, BERT)	0 (Free)
Cloud Services	Google Colab (Free Tier) & Render/Railway Free Tier	0 (Free)
Connectivity	High-speed Internet for Model Training & Research	15,000
Software/Web	Domain Name (.com/.lk) & Basic Hosting (Annual)	8,000
Documentation	Printing, Binding, and Thesis Preparation	7,000
Total		30,000

Table 5 - Estimated Budget Breakdown

7.2 Detailed Justification for Each Expense

7.2.1 Hardware Costs

- **Justification:** As this is an academic research project, the primary hardware used will be the our existing personal computer.

7.2.2 Software Licenses

- **Justification:** All software frameworks, including Python, FastAPI, and BERT, are Open Source, meaning there are no licensing costs involved.[19]

7.2.3 Cloud Services

- **Justification:** To train the BERT and LSTM models, Google Colab **(**Free Tier) [20] will be utilized, providing access to high-performance GPUs at no cost. For the final web application hosting, platforms with a free entry tier (e.g., Vercel) will be used to demonstrate the prototype.

7.2.4 Connectivity and Research Access

- **Justification:** Consistent high-speed internet is essential for downloading large pre-trained models (BERT-base)[19] and accessing IEEE/ACM research databases. This budget covers the specialized data requirements for the project duration

7.2.5 Web Hosting and Domain

Justification: To evaluate the User Requirements, a live web URL is needed. A nominal fee is allocated for a professional domain and basic web hosting to make the system accessible to stakeholders for testing.

8. WORK BREAKDOWN STRUCTURE (WBS)

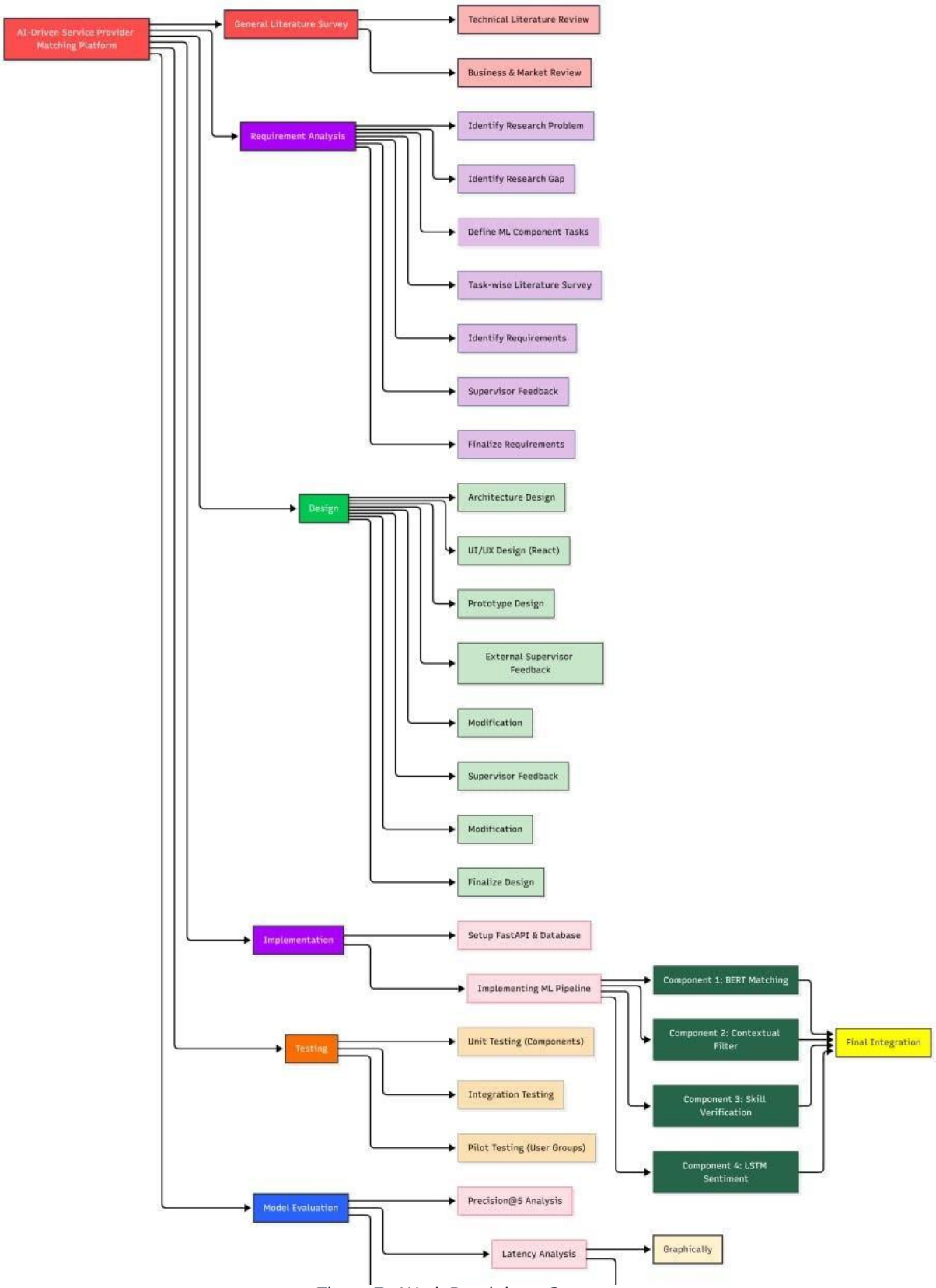


Figure 7 - Work Breakdown Structure

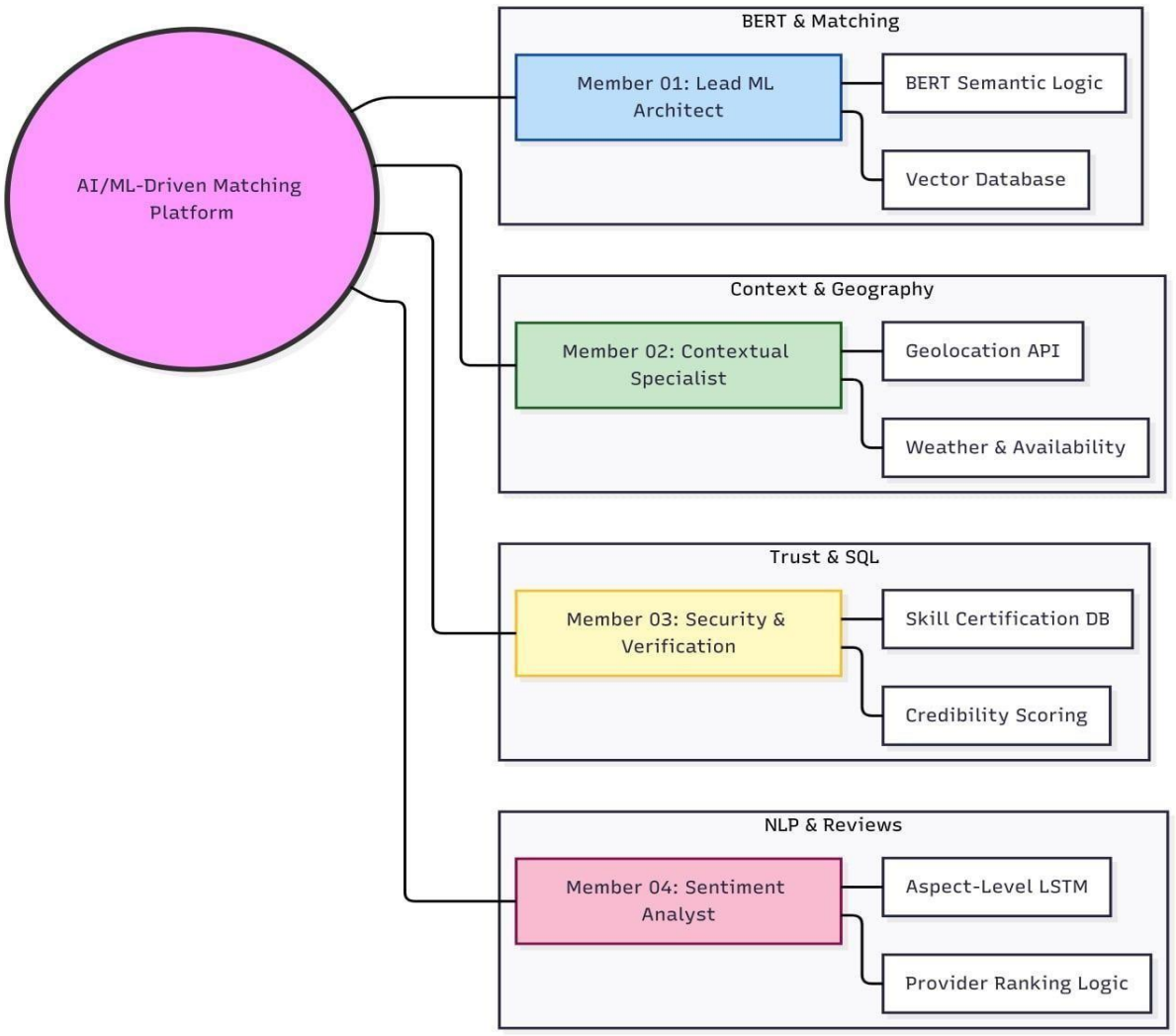


Figure 8 - Member-wise Work Breakdown Structure

Phase	Task ID	Task Name	Description	Deliverable	Responsible
Planning	T1	Requirement Analysis	Identify system needs	Requirement document	Project team
Planning	T2	Feasibility Study	Assess viability	Feasibility report	Project manager
Design	T3	System Architecture	Define architecture	Architecture diagram	System architect
Design	T4	Database Design	Create database schema	ER diagram	Database engineer
Development	T5	Backend Development	Implement APIs	Backend system	Developers
Development	T6	Mobile Development	Build mobile interface	Mobile app	Mobile developers
Testing	T7	Unit Testing	Test modules	Test results	QA team

Testing	T8	System Testing	Full system validation	Test report	QA team
Deployment	T9	System Deployment	Deploy to cloud	Live system	DevOps

Table 6 - Work Breakdown Structure table

9. GANTT CHART

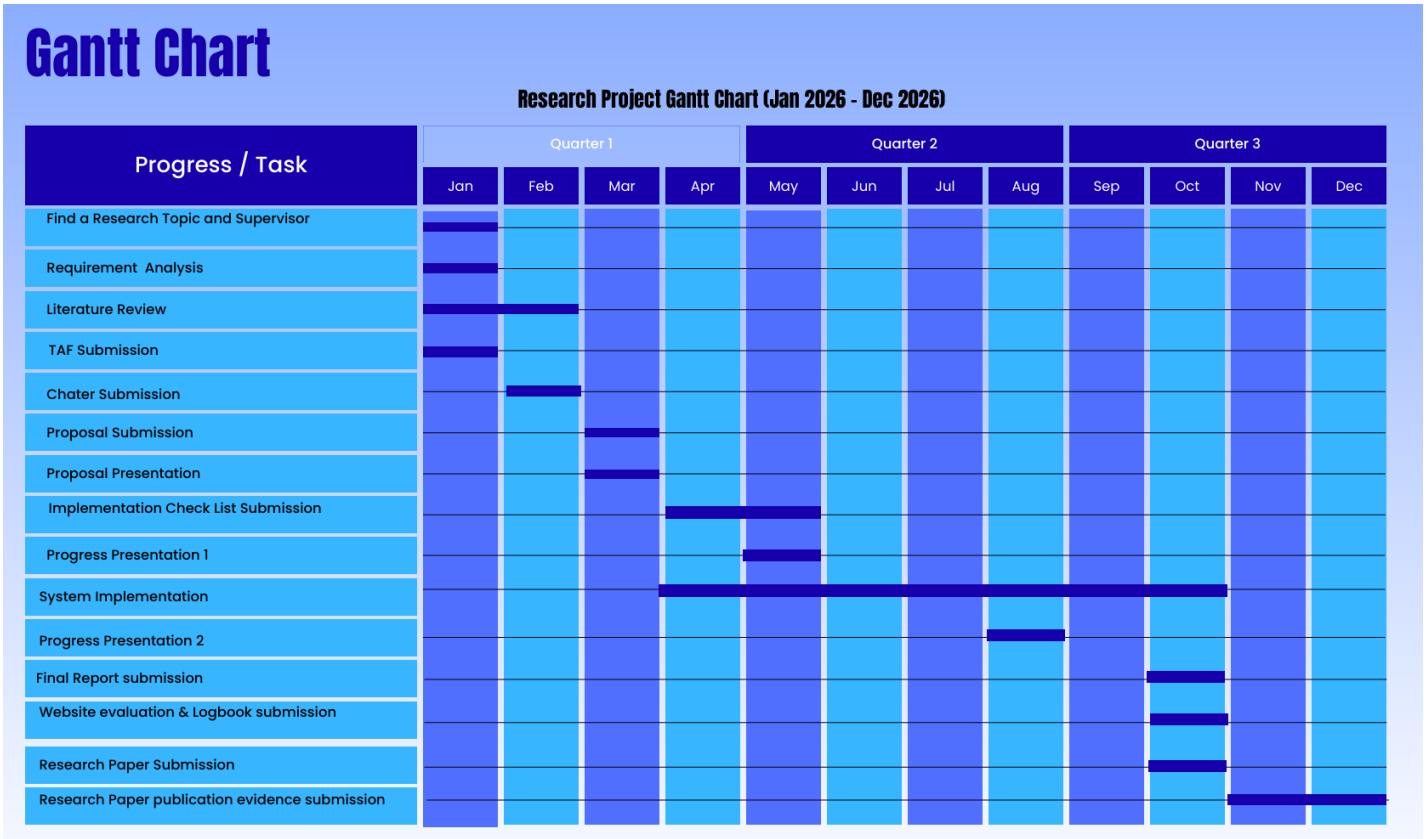


Figure 9 - Gantt Chart

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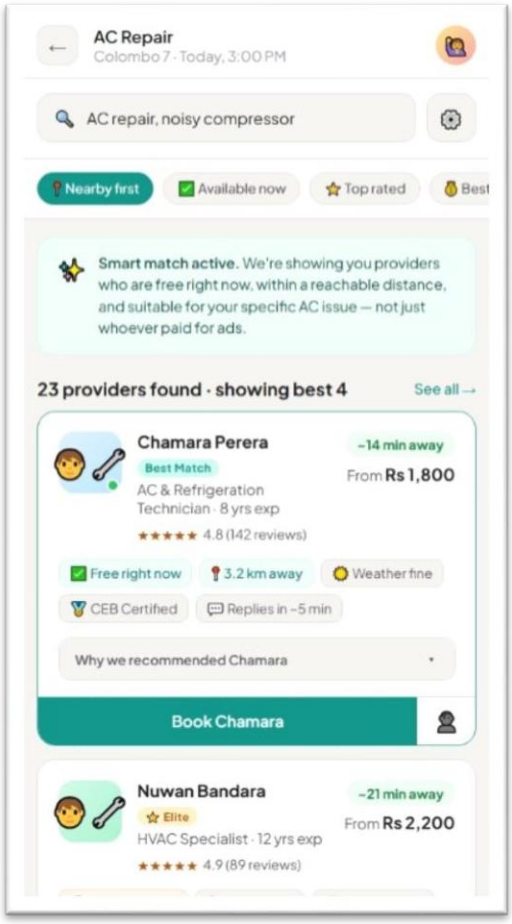
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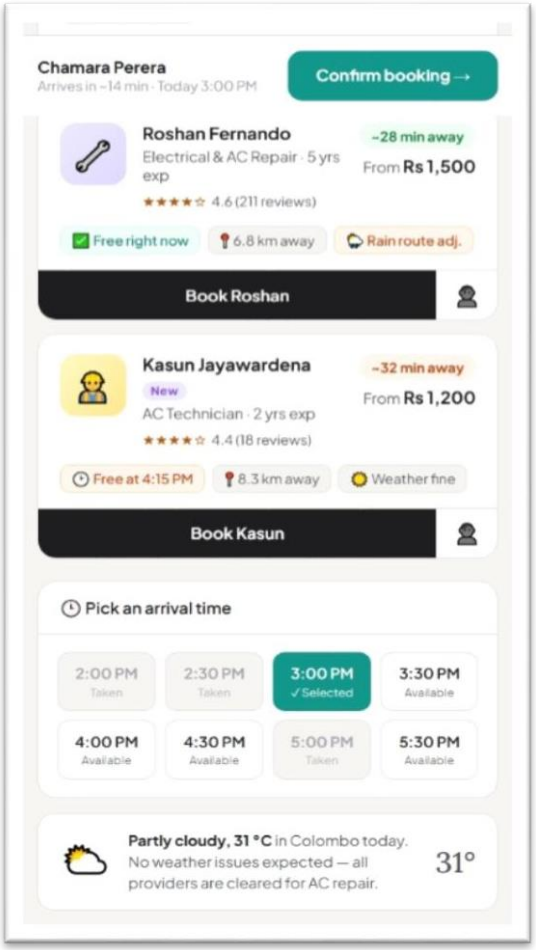
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11. APPENDICES

11.1 UI Mockups





How we score each provider

Four real-time signals, weighted by importance

40% — **Distance & travel time**
Haversine distance + Google Maps live traffic to get real ETA, not just straight-line km.

30% — **Availability right now**
ARIMA time-series forecast + live calendar conflict detection across provider schedules.

20% — **Weather compatibility**
Matches service types to live weather — e.g. heavy rain deprioritises outdoor electrical work.

10% — **Demand & peak hours**
Accounts for provider overload at peak times — redirects demand to available alternatives.

The matching pipeline

Component 2 sits between recommendation and credibility scoring

STEP 01 — **Input from Component 1**
Receives top-20 providers ranked by hybrid CF + CBF + BERT recommendation engine.
20 candidates in

STEP 02 — **Distance filter**
Haversine formula + Google Maps Distance Matrix API for accurate travel time per provider.
Proximity ranked

STEP 03 — **Availability check**
ARIMA forecast + WebSocket real-time status + schedule conflict detection engine.
Feasible slots only

STEP 04 — **Context scoring**
Weather matrix + demand index applied as weighted multipliers on each provider score.
Final score assigned

OUTPUT — **Top 10 passed on**
Contextually feasible top-10 sent to Component 3 for skill verification + credibility tiering.
10 providers out

Matched providers for you

23 nearby · filtered to 10 · showing top 4

1 **Chamara Perera** Trusted
3.2 km away · Free right now · Good weather · 4.8 · 142 jobs
94 match score
~14 min ETA

2 **Nuwan Bandara Silva** Elite
5.1 km away · Free in 45 min · Good weather · 4.9 · 89 jobs
91 match score
~21 min ETA

3 **Roshan Fernando** Available
6.8 km away · Free right now · Route adjusted · 4.6 · 211 jobs
88 match score
~28 min ETA

4 **Kasun Jayawardena**
8.3 km away · Free at 4:15 PM · Good weather · 4.7 · 67 jobs
85 match score
~32 min ETA

Weather now · Colombo
31 °C · Partly cloudy
Humidity 72% · Wind 14 km/h NE
OpenWeatherMap · updated 3 min ago
No weather restrictions for AC repair today

AC repair demand today
Peak at 3 PM — 23 providers active in Colombo right now

Live data sources
- Google Maps API — 98 ms
- OpenWeatherMap — 124 ms
- Redis availability cache — 12 ms
- CEB Registry (mock) — 214 ms
- WebSocket tracker — Live

11.2 Risk Assessments and Mitigation

Risk Category	Risk Description	Probability	Impact	Risk Score (0-10)	Mitigation Strategy
Technical	BERT/LSTM models fail to meet accuracy targets (85%) due to bilingual nuances.	Medium	High	6	Fine-tune models with a local bilingual corpus; use Attention Mechanisms to focus on key intent.
Data	Limited availability of historical booking data for Collaborative Filtering.	High	High	9	Use Hybrid Fusion to give more weight to BERT (35%) and CBF (30%) until data grows.
Security	Leakage of sensitive user data (geolocation, contact details, payment info)	Low	Very High	8	Implement AES encryption, strict access control policies, and secure database schemas.
Performance	Total pipeline latency exceeds 10 seconds per request.	Medium	Medium	4	Implement parallel processing for CF, CBF, and BERT and use caching for weather/location data.
Business / Adoption	Users or providers resist adoption due to lack of trust in AI-driven matching.	Medium	Medium	4	Provide transparent explanations for recommendations; ensure a bilingual (Sinhala/English) user-friendly UI.
External	Dependency on external Weather/Map APIs leads to system failures during outages.	Medium	High	4	Enable offline fallback logic; cache the most recent location and weather data locally for 2 hours.
Resource	Unavailability of expert technicians to verify the	Low	High	5	Collaborate with vocational training institutes (e.g., NAITA) to

11.3 Plagiarism Report

